



K. S. INSTITUTE OF TECHNOLOGY

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No. 14, Raghuvanahalli, Kanakapura Road, Bengaluru – 560109

Department of Computer Science and Engineering

Blockchain Technology(25MCS214A)

Topic: Secure Electronic Health Records

Presented By:

B Nayana,

1KS25SCS01

Semester: 2 Mtech,

K S Institute of Technology.

Guided By:

Dr.Sunita Chalageri,

Professor,

K S Institute of Technology –

Bangalore.

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Introduction

- Centralized systems face issues like data breaches, poor interoperability, and limited patient control.
- Growing healthcare data requires secure, reliable, and tamper-proof data management.
- Blockchain provides decentralized, immutable, and transparent data storage for EHR systems.
- Ensures secure access, encryption, audit tracking, and gives patients control over their health data.

Objectives

- To design and develop a secure Electronic Health Record (EHR) management system using blockchain technology to ensure data integrity and confidentiality.
- To eliminate the limitations of traditional centralized EHR systems by implementing a decentralized and tamper-proof data storage mechanism.
- To provide role-based access control so that only authorized users such as doctors, patients, and administrators can access specific medical records.
- To enable patients to have full control over their health data, including the ability to grant and revoke access permissions.
- To ensure secure data sharing among healthcare providers while maintaining privacy through encryption techniques.

System Architecture

- Patient provides data through a secure data input system
- Healthcare provider receives and accesses patient records
- Data is transmitted to a blockchain network for secure processing
- Blockchain ensures data integrity and tamper-proof storage
- Verified data is stored in a centralized/decentralized database
- Enables secure sharing of medical records
- Improves transparency, trust, and data security
- Reduces risk of data manipulation and unauthorized access

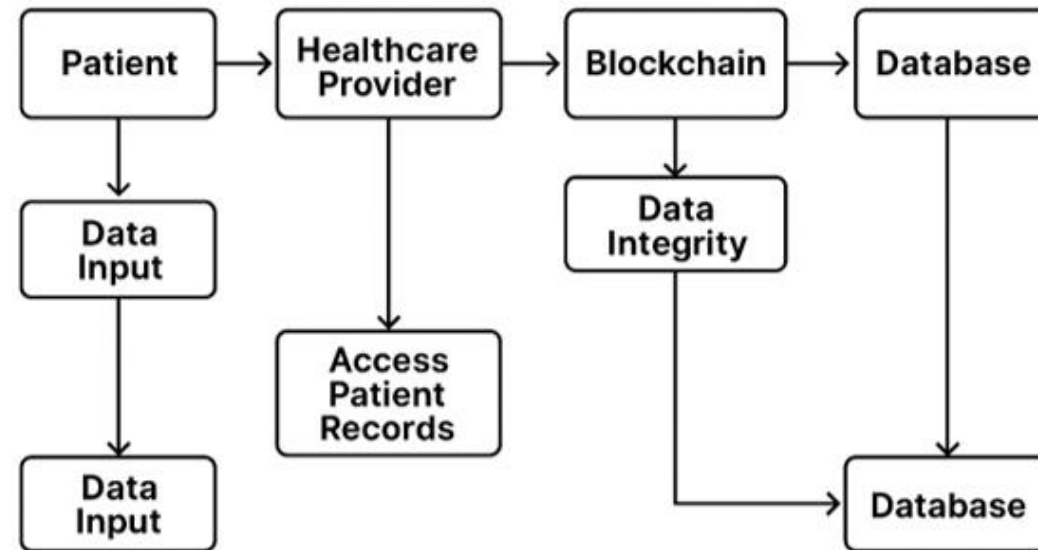


Fig 1: System architecture of Secure health records

Proposed Methodology

- Collect patient health records from authorized healthcare providers.
- Encrypt all medical data to ensure confidentiality.
- Implement secure user authentication and access control.
- Store records in a secure cloud/database environment.
- Enable authorized sharing of records between healthcare institutions.
- Maintain audit logs to track all access and modifications.
- Provide patients with controlled access to their health records.
- Continuously monitor the system for security threats and data breaches.

Expected Outcome

- Improved data security
- Enhanced patient privacy
- Reduced risk of data breaches
- Faster and secure access to medical records
- Better healthcare data management

Workflow

**Data Collection → Encryption → Secure Storage → Authentication → Authorized Access
→ Audit Monitoring → Secure Sharing**

Implementation

```
class Blockchain:
    def __init__(self):
        self.chain = [self.create_genesis_block()]

    def create_genesis_block(self):
        return Block(0, {"msg": "Genesis Block"}, "0")

    def add_block(self, data):
        prev_block = self.chain[-1]
        new_block = Block(len(self.chain), data,
prev_block.hash)
        self.chain.append(new_block)

    def display_chain(self):
        print("\n===== BLOCKCHAIN LEDGER
=====")
        for block in self.chain:
            print(f"\nBlock Index: {block.index}")
            print(f"Time:
{time.ctime(block.timestamp)}")
            print(f>Data: {block.data}")
            print(f"Hash: {block.hash[:25]}...")
            print(f"Prev Hash:
{block.previous_hash[:25]}...")
```

```
if __name__ == "__main__":
    system = EHRSystem()
    # Register users
    patient1 =
system.register_patient("Nayana")
    doctor1 = system.register_doctor("Dr.
Sharma")
    # Grant access
    system.grant_access(patient1, doctor1)

    # Add records
    system.add_record(
        doctor1, patient1,
        "Fever",
        "Paracetamol 500mg"
    )
    system.add_record(
        doctor1, patient1,
        "Cold",
        "Antibiotics"
    )
    # View records
    system.view_records(doctor1, patient1)
```

```
# Revoke access
system.revoke_access(patient1, doctor1)

# Try adding record after
revoke
system.add_record(
    doctor1, patient1,
    "Follow-up",
    "Rest"
)
```

Results and Outputs

Block 1 @ Tue Jul 8 15:27:12 2025

Data: {'patient_id': 'PAT_98xy76pq', 'name': 'Nayana', 'action': 'REGISTER_PATIENT'}

Hash: a1f5c3ab879281bb72f0... Prev: 00000000000000000000...

Block 2 @ Tue Jul 8 15:27:13 2025

Data: {'doctor_id': 'DOC_12ab34cd', 'name': 'Dr. Sharma', 'action': 'REGISTER_DOCTOR'}

Hash: b21cbe37bb717aa91f3d... Prev: a1f5c3ab879281bb72f0...

Block 3 @ Tue Jul 8 15:27:14 2025

Data: {'action': 'GRANT_ACCESS', 'patient': 'PAT_98xy76pq', 'doctor': 'DOC_12ab34cd'}

Hash: c31de98ab8712cc345aa... Prev: b21cbe37bb717aa91f3d...

Block 4 @ Tue Jul 8 15:27:15 2025

Data: {'record_id': 'REC_a1b2c3', 'diagnosis': 'Fever', 'treatment': 'Paracetamol', 'status': 'ADDED'} Hash:

d41ac78bb91234ef78bd... Prev: c31de98ab8712cc345aa...

Block 5 @ Tue Jul 8 15:27:16 2025

Data: {'record_id': 'REC_d4e5f6', 'diagnosis': 'Cold', 'treatment': 'Antibiotics', 'status': 'ADDED'} Hash:

e59fd23ab67891bc2345... Prev: d41ac78bb91234ef78bd...

Conclusion

- The Secure Electronic Health Record (EHR) Management System using Blockchain successfully demonstrates a modern approach to handling sensitive healthcare data with enhanced security, transparency, and reliability
- The implementation of blockchain provides immutability, meaning that once a record is created, it cannot be altered without proper authorization. This guarantees data integrity and prevents unauthorized modifications
- The hybrid storage model, combining blockchain for verification and database systems for performance, provides an optimal balance between security and efficiency